🌱 Sustainable Tomorrow

☀️ Light ≡

🤖

Advanced AI analysis of three climate scenarios through 2050, including the impact of recent policy changes and political developments on global climate action.

2.4°C	1.5°C	1.8°C	15%
Current Trends by 2050	Aggressive Action Target	Paris Agreement Path	Aggressive Action Probability
Catastrophic impacts likely	Still achievable with effort	Above 1.5°C target	Requires unprecedented cooperation

🎛️ Scenario Controls

Climate Scenario

🌍 All Scenarios

📈 Current Trends

🤝 Paris Agreement

🚀 Aggressive Action

Time Horizon

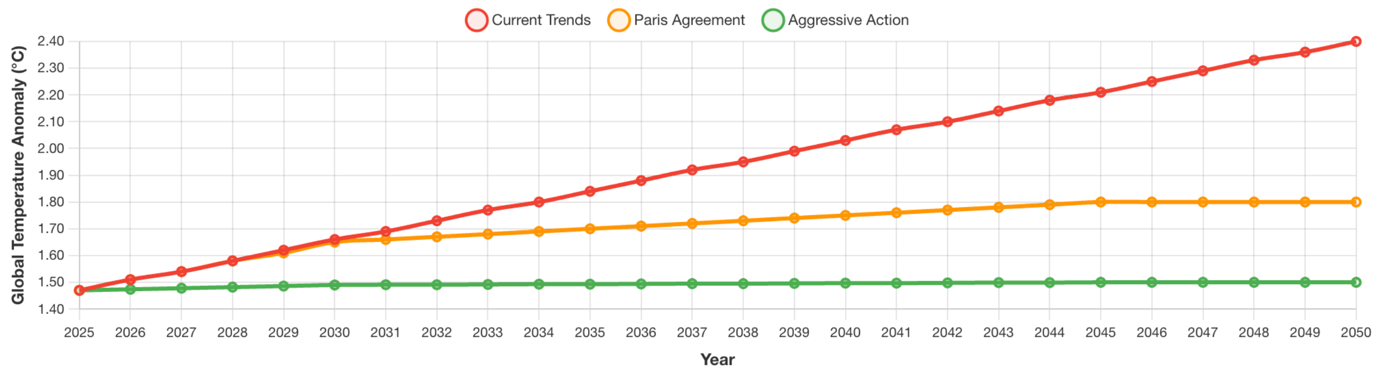
📅 Near-term (2030)

🔮 Long-term (2050)

Temperature Projections by Scenario (2025-2050)

AI Model: Ensemble of CMIP6 climate models with policy impact analysis
Scenarios: All three scenarios compared
Key Finding: Policy choices in the next 5 years determine our climate future

🤖 Sustainability Chatbot



🔍 AI Analysis

The gap between scenarios widens dramatically after 2030, showing how current policy decisions will determine our climate future. Recent AI models show improved accuracy using real-time satellite data.

📊 Select a Specific Scenario

Choose **Current Trends**, **Paris Agreement**, or **Aggressive Action** above to view detailed predictions including renewable energy growth and CO_2 emissions trajectory.

About This Project

Created for the **Viz Con 2025** contest at **Analyticon**(Amazon-wide Conference on Data Engineering, Science and Analytics). This interactive application explores the connections between human activity and environmental conservation through compelling data narratives using real datasets.

Data Sources

🌡️ **Temperature Data:**
NASA GISTEMP v4 (Goddard Institute)
⚡ **Renewable Energy:**
IRENA Global Energy Statistics
⛏️ **Critical Minerals:**
IEA Critical Minerals Data Explorer
🧊 **Sea Ice Data:**
NSIDC Sea Ice Index
🏭 **CO2 Data:**
NOAA Mauna Loa Observatory

Technology & AI

Frontend: React.js, D3.js, Chart.js, Framer Motion
Backend: Python Flask, NumPy, Pandas
AI Integration: Pattern recognition, correlation analysis, predictive modeling, and natural language insight generation
AI Chatbot: Amazon Bedrock (Nova Pro Model), API Gateway, AWS Lambda, S3, IAM
Deployment: AWS Amplify with automated CI/CD pipeline

Viz Con 2025 Entry

Entry for the **2025 Viz Con contest**, part of the **Analyticon** - Amazon-wide Conference on Data Engineering, Science and Analytics
Challenge: "Visualizing a Sustainable Tomorrow"
Approach: Multi-dataset integration with real NASA GISTEMP, NOAA, IRENA, and IEA data
Goal: Inspire climate action through compelling data storytelling
Features: Interactive visualizations, correlation analysis, predictive modeling, and accessibility-first design

Built with 💚 by @ygal and his best friend Amazon Q 🤖

Powered by AWS | NASA, IRENA, IEA, NSIDC & NOAA data